Time Series Data Analysis

Problem statement: Analysis and Visualization of Stock Market Data Dataset: "Stock_Prices.csv" Description: The dataset contains historical stock price data for a particular company over a period of time. It includes attributes such as date, closing price, volume, and other relevant features. The goal is to perform time series data analysis on the stock price data to identify trends, patterns, and potential predictors, as well as build models to forecast future stock prices. Tasks to Perform:

1. Import the "Stock_Prices.csv" dataset.
2. Explore the dataset to understand its structure and content.
3. Ensure that the date column is in the appropriate format (e.g., datetime) for time series analysis.
4. Plot line charts or time series plots to visualize the historical stock price trends over time.
5. Calculate and plot moving averages or rolling averages to identify the underlying trends and smooth out noise.
6. Perform seasonality analysis to identify periodic patterns in the stock prices, such as weekly, monthly, or yearly fluctuations.
7. Analyze and plot the correlation between the stock prices and other variables, such as trading volume or market indices.
8. Use autoregressive integrated moving average (ARIMA) models or exponential smoothing models to forecast future stock prices.

Code :

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.tsa.seasonal import seasonal_decompose  # <-- Add this line
from statsmodels.tsa.arima.model import ARIMA
from sklearn.metrics import mean_squared_error
import warnings
warnings.filterwarnings('ignore')

df = pd.read_csv('NSE-Tata_Global_Beverages_Limited.csv')
```

2.Explore the dataset to understand its structure and content.

```
df.head()
```

	Date	Open	High	Low	Last	Close	Total Trade Quantity	Turnover (Lacs)
0	2018-10-08	208.00	222.25	206.85	216.00	215.15	4642146.0	10062.83
1	2018-10-05	217.00	218.60	205.90	210.25	209.20	3519515.0	7407.06
2	2018-10-04	223.50	227.80	216.15	217.25	218.20	1728786.0	3815.79
3	2018-10-03	230.00	237.50	225.75	226.45	227.60	1708590.0	3960.27
4	2018-10-01	234.55	234.60	221.05	230.30	230.90	1534749.0	3486.05

df.shape

(1235, 8)

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1235 entries, 0 to 1234
Data columns (total 8 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Date                  1235 non-null   object
 1   Open                  1235 non-null   float64
 2   High                  1235 non-null   float64
 3   Low                   1235 non-null   float64
 4   Last                  1235 non-null   float64
 5   Close                 1235 non-null   float64
 6   Total Trade Quantity  1235 non-null   float64
 7   Turnover (Lacs)       1235 non-null   float64
dtypes: float64(7), object(1)
memory usage: 77.3+ KB
```

df.describe()

	Open	High	Low	Last	Close	Total Trade Quantity	Turnover (Lacs)
count	1235.000000	1235.000000	1235.000000	1235.000000	1235.000000	1.235000e+03	1235.000000
mean	168.954858	171.429069	166.402308	168.736356	168.731053	2.604151e+06	4843.166502
std	51.499145	52.436761	50.542919	51.587384	51.544928	2.277028e+06	5348.919832
min	103.000000	104.600000	100.000000	102.600000	102.650000	1.001800e+05	128.040000
25%	137.550000	138.925000	135.250000	137.175000	137.225000	1.284482e+06	1801.035000
50%	151.500000	153.250000	149.500000	151.200000	151.100000	1.964885e+06	3068.510000
75%	169.000000	172.325000	166.700000	169.100000	169.500000	3.095788e+06	5852.600000
max	327.700000	328.750000	321.650000	325.950000	325.750000	2.919102e+07	55755.080000

df.isnull().sum()

```
Date                    0
Open                    0
High                    0
Low                     0
Last                    0
Close                   0
Total Trade Quantity    0
Turnover (Lacs)         0
dtype: int64
```

3. Ensure that the date column is in the appropriate format (e.g., datetime) for time series analysis.

```
df['Date'] = pd.to_datetime(df['Date'])
df.set_index('Date', inplace=True)
```

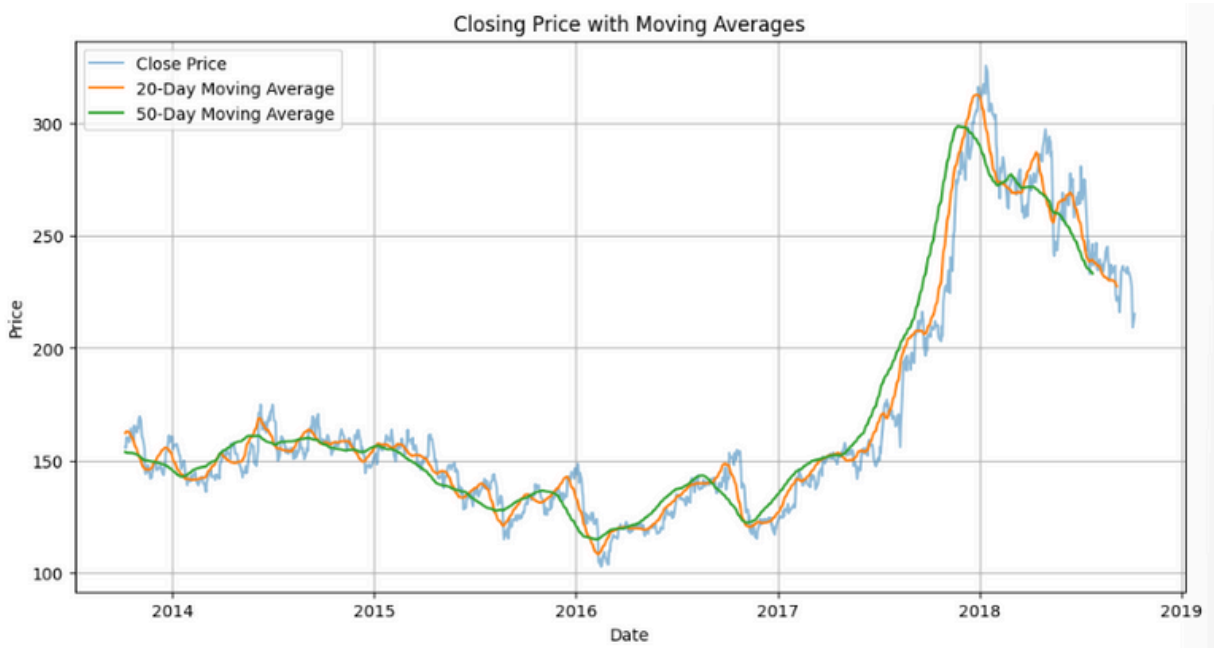
4. Plot line charts or time series plots to visualize the historical stock price trends over time.

```
import matplotlib.pyplot as plt

plt.figure(figsize=(12,6))
plt.plot(df['Close'], label='Close Price')
plt.title('Historical Closing Price')
plt.xlabel('Date')
plt.ylabel('Price')
plt.legend()
plt.grid(True)
plt.show()
```

5. Calculate and plot moving averages or rolling averages to identify the underlying trends and smooth out noise.

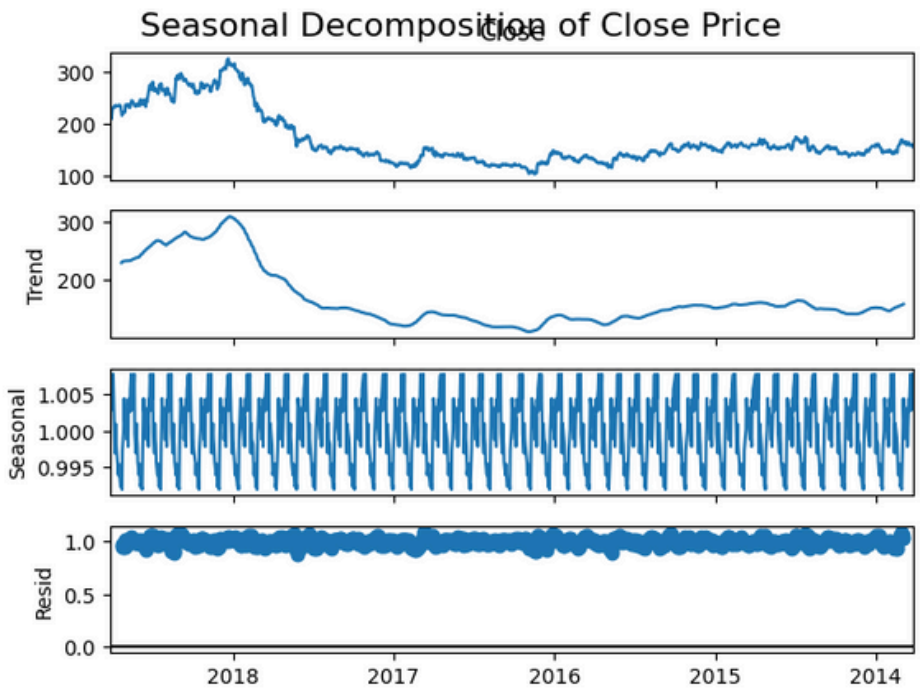
```
df['MA_20'] = df['Close'].rolling(window=20).mean()
df['MA_50'] = df['Close'].rolling(window=50).mean()

plt.figure(figsize=(12,6))
plt.plot(df['Close'], label='Close Price', alpha=0.5)
plt.plot(df['MA_20'], label='20-Day Moving Average')
plt.plot(df['MA_50'], label='50-Day Moving Average')
plt.title('Closing Price with Moving Averages')
```

```
plt.xlabel('Date')
plt.ylabel('Price')
plt.legend()
plt.grid(True)
plt.show()
```

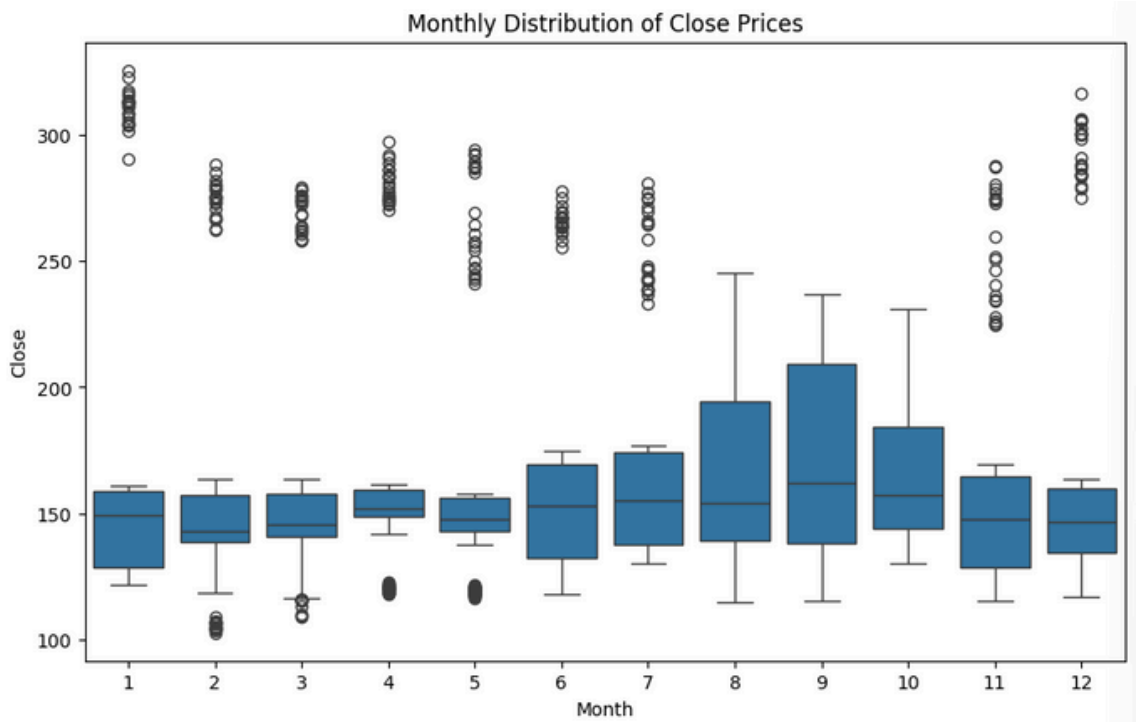


6. Perform seasonality analysis to identify periodic patterns in the stock prices, such as weekly, monthly, or yearly fluctuations.

```
from statsmodels.tsa.seasonal import seasonal_decompose

result = seasonal_decompose(df['Close'].dropna(), model='multiplicative', period=30)
result.plot()
plt.suptitle('Seasonal Decomposition of Close Price', fontsize=16)
plt.show()
```



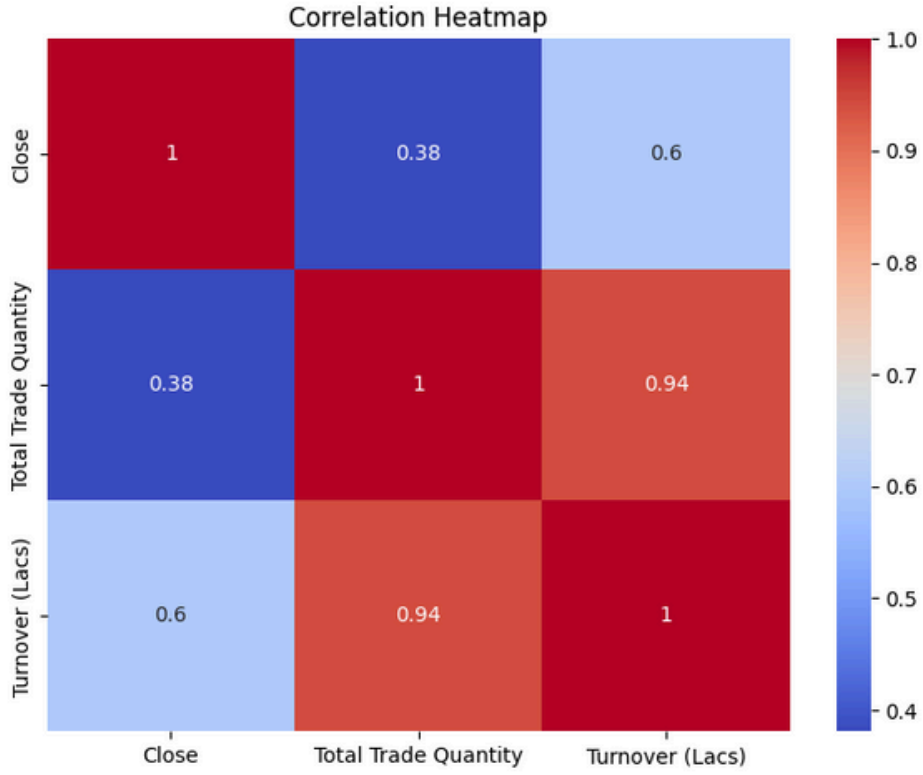
```
df['Month'] = df.index.month
plt.figure(figsize=(10,6))
sns.boxplot(x='Month', y='Close', data=df)
plt.title('Monthly Distribution of Close Prices')
plt.show()
```



7. Analyze and plot the correlation between the stock prices and other variables, such as trading volume or market indices.

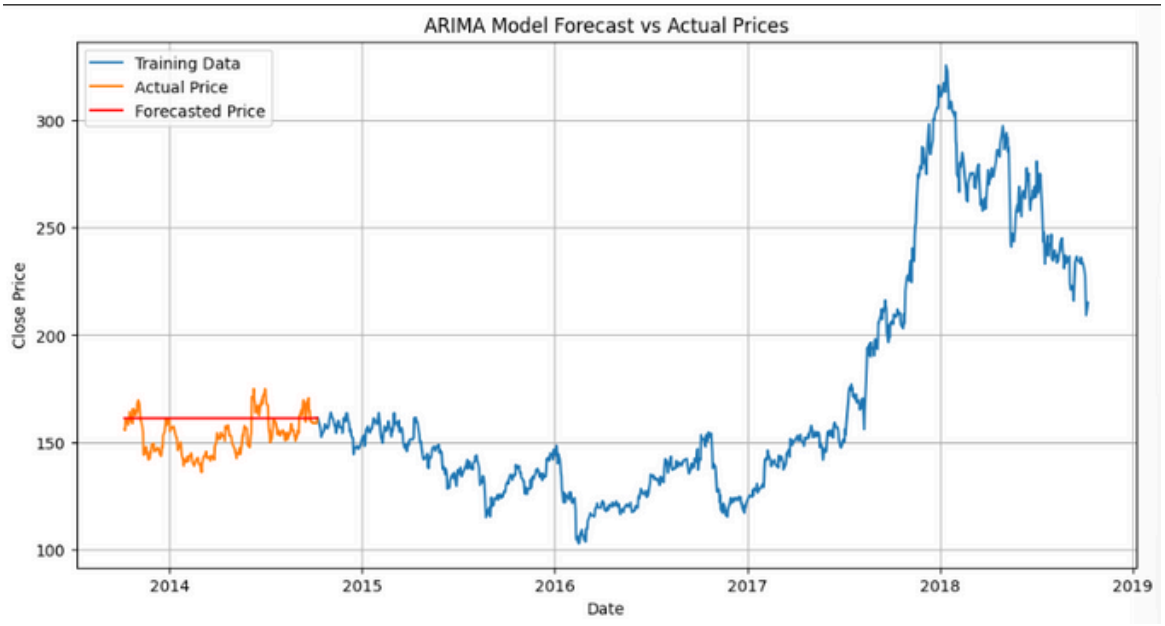
```
orr_df = df[['Close', 'Total Trade Quantity', 'Turnover (Lacs)']]
print("\nCorrelation matrix:")
print(corr_df.corr())

plt.figure(figsize=(8,6))
sns.heatmap(corr_df.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

```
Correlation matrix:
                      Close  Total Trade Quantity  Turnover (Lacs)
Close              1.000000              0.380801         0.599155
Total Trade Quantity 0.380801            1.000000         0.941976
Turnover (Lacs)    0.599155              0.941976         1.000000
```



8. Use autoregressive integrated moving average (ARIMA) models or exponential smoothing models to forecast future stock prices.

```
# Prepare data
data = df['Close'].dropna()

# Split train-test
train_size = int(len(data)*0.8)
train, test = data[:train_size], data[train_size:]

# Fit ARIMA model (order can be adjusted)
model = ARIMA(train, order=(5,1,0))
model_fit = model.fit()

# Forecast
forecast = model_fit.forecast(steps=len(test))

# Evaluate
mse = mean_squared_error(test, forecast)
print(f"\nMean Squared Error (ARIMA Forecast): {mse:.4f}")
```

```
Mean Squared Error (ARIMA Forecast): 135.0314
```

```
# Plot forecast vs actual
plt.figure(figsize=(12,6))
plt.plot(train.index, train, label='Training Data')
plt.plot(test.index, test, label='Actual Price')
plt.plot(test.index, forecast, label='Forecasted Price', color='red')
plt.title('ARIMA Model Forecast vs Actual Prices')
plt.xlabel('Date')
plt.ylabel('Close Price')
plt.legend()
plt.grid(True)
plt.show()
```